

Risk-Based Operation Coordination of Water Desalination and Renewable-Rich Power Systems

Mohamed Elsir^{ID}, Ali T. Al-Awami^{ID}, *Senior Member, IEEE*, Mohamed A. Antar, Konstantinos Oikonomou, *Member, IEEE*, and Masood Parvania^{ID}, *Senior Member, IEEE*

Abstract—Handling the variability and uncertainty associated with integrating large capacities of renewable energy sources (RES) into the power grid is a challenge that is increasingly influencing the power systems operation. At the same time, the growing need for desalinated water in arid areas increases the importance of suitable energy sources for sustainable operation of water desalination plants. However, as power and water system operators have traditionally operated their systems in isolation, there is a lack of understanding of the interdependence and interactions between these two systems. This paper addresses this gap by proposing a risk-based two-stage stochastic co-optimization framework that coordinates the operation of a renewable-rich power system with the operation of grid-connected reverse-osmosis water desalination plants (RO-WDP) to minimize their combined operational costs while increasing the utilization of RES. From the power system operation standpoint, the RO-WDPs are considered as controllable demand, and the proposed model integrates the energy flexibility of RO-WDPs in the day-ahead power system operation. The proposed model considers the operational constraints of both power and water desalination systems, thus co-optimizing their operation without compromising the reliable supply of power and water to end-users, while taking into account the uncertainty of the demands and RES. Simulation results demonstrate the benefits of the proposed coordination on enhancing the power system efficiency, facilitating RES integration, and minimizing the combined operational costs of both systems while minimizing their operating risk using conditional value at risk.

Index Terms—Energy-water nexus, energy flexibility, risk-based operation, water desalination plants.

Manuscript received 1 July 2021; revised 19 November 2021 and 28 February 2022; accepted 19 April 2022. Date of publication 12 May 2022; date of current version 27 February 2023. This work was supported in part by the Interdisciplinary Research Center of Smart Mobility & Logistics, King Fahd University of Petroleum & Minerals, under Grant DF201006. Paper no. TPWRS-01045-2021. (Corresponding author: Masood Parvania.)

Mohamed Elsir is with the Department of Electrical Engineering, King Fahd University of Petroleum & Minerals, Dhahran 31261, Saudi Arabia (e-mail: m.elsir@hotmail.com).

Ali T. Al-Awami is with the Department of Electrical Engineering and Interdisciplinary Research Center for Smart Mobility & Logistics, King Fahd University of Petroleum & Minerals, Dhahran 31261, Saudi Arabia (e-mail: aliawami@kfupm.edu.sa).

Mohamed A. Antar is with the Department of Mechanical Engineering and the Interdisciplinary Research Center for Membranes and Water Security, King Fahd University of Petroleum & Minerals, Dhahran 31261, Saudi Arabia (e-mail: antar@kfupm.edu.sa).

Konstantinos Oikonomou is with the Pacific Northwest National Laboratory, Richland, WA 99354 USA (e-mail: konstantinos.oikonomou@pnl.gov).

Masood Parvania is with the Department of Electrical and Computer Engineering, University of Utah, Salt Lake City, UT 84112 USA (e-mail: masood.parvania@utah.edu).

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TPWRS.2022.3174565>.

Digital Object Identifier 10.1109/TPWRS.2022.3174565

NOMENCLATURE

Subscripts, Superscripts, and Indices

D	Subscript of the uncontrolled power demands.
d	Superscript of wind generation plants.
$f/b/p$	Superscript of feed/brine/permeate water side.
g	Index of power generation unit; $g = 1, 2, \dots, N^{th}$
i	The index that corresponds to ψ^w segmentation; $i = 1, 2, \dots, n^w - 1$
j	The index that corresponds to ω^w segmentation; $j = 1, 2, \dots, m^w - 1$
k	Index of WDP/WST units; $k = 1, 2, \dots, N^{WDP}$
pv	Index of PV generator plant; $pv = 1, 2, \dots, N^v$
s	Index of scenario; $s = 1, 2, \dots, N^s$
t	Index of time; $t = 1, 2, \dots, N^t$
th	Superscript of thermal generation plants.
v	Superscript of PV generation plants.
w	Superscript of WDP/WST units.
wi	Index of Wind generator plant; $wi = 1, 2, \dots, N^d$
z/z'	Index for sending/receiving buses.

Parameters

α_1/α_2	Temperature correction parameters (K^{-1})
$\bar{P}/\bar{P}$	Min/Max power production limit (MW)
$\bar{Q}/\bar{Q}$	Min/Max water flow rate (m^3/hr)
$\bar{U}_p/\bar{D}_n$	Minimum up/down time for thermal generators (hr)
Δt	Time interval (= 1 hr)
η^{Pump}	High pressure pump efficiency.
κ	Proportional factor to represent the relation between the osmotic pressure and water salinity.
λ	The spinning reserve factor.
μ	Limit parameters for a WST.
ω	The associated segments for Q_f
ψ	The associated segments for P_f
σ	A limiting percentage of applied hydraulic pressure.
τ	The initial water volume percentage.
$\tilde{P}$	RES forecasted p.u. power.
Ω	Probability of each scenario.
φ	Risk aversion parameter.
ϱ	The level of confidence.
b	Operation cost coefficient (\$/MW)
C^F	Concentration polarization factor.
C_{SU}/C_{SD}	Start-up/shut-down cost.
D^w	Water demand (m^3/hr)

K^{wat}/K^s	water/salt membrane permeability coefficients.
L^D	Uncontrolled electric load (MW)
N_m	The number of RO membrane modules.
Re	Percent volume of feedwater recovered as permeate.
RU/RD	Maximum production ramp up/down rate.
S_e	Surface area of RO membrane (m^2)
SR	Available spinning reserve (MW)
T	Water temperature ($^{\circ}C$)
T^F	The temperature correction factor (K)
Up_o/Dn_o	Initial up/down time for thermal generators (hr)
V_o	The initial water volume in a WST (m^3)
VK	WST capacity limit (m^3)
x	Line reactance (Ω)

Decision Variables

$\bar{X}$	The average dissolved salt concentration (kg/m^3)
β	Weights $\in [0, 1]$ to calculate the piece-wise approximated function inside the concerned intervals.
$\Delta\bar{H}/\Delta\bar{\pi}$	Mean applied hydraulic/osmotic pressure difference.
γ	Participation factor of a WDP to the water demand.
$\hat{P}$	Curtailed power of RES (MW)
$\bar{h}/\underline{h}$	Upper/lower binary variables of the rectangle heads.
ρ	A positive auxiliary variable for computing $CVaR$
θ	Bus voltage angles (degree)
$CVaR$	The conditional value at risk.
H	RO water applied hydraulic pressure (kPa)
$H^{f,ap}$	The piece-wise approximation of P^f
$L^{w,ap}$	The piece-wise approximation of L^w
L^w	Energy consumption for a WDP (MW/ m^3)
M^s	The salt transport across RO membrane (kg/hr)
P	Scheduled thermal and RES generation (MW)
P^L	Active power flow (MW)
Q	Water flow rate (m^3/hr)
Q^{WST}	Water flow from a WST (m^3/hr)
$Q_{f,ap}$	The piece-wise approximation of Q^f
SU/SD	Startup/shut-down indicators.
u	Unit status.
V^{WST}	The water volume in a WST (m^3)
VaR	The value at risk.
X	Water dissolved salt concentration (kg/ m^3)

I. INTRODUCTION

A. Background and Motivation

THE water-supply system (WSS) is a critical infrastructure that is crucial for satisfying the domestic, industrial, and agricultural water needs of any civilized community. The WSS consists of two major segments: the water distribution system (WDS), which includes a network of pipelines, pumps, and valves, and the water resources, which consist of water reservoirs and water desalination plants (WDPs). Evidently, a functioning WSS requires energy. Likewise, producing energy requires water. This inter-relationship is often called energy-water nexus,

a term that refers broadly to the inter-dependencies among power and water systems [1]. Recently, increased attention has been given to understating this interdependency, including the potential to use controllable loads from water facilities to provide a range of services (e.g., energy flexibility, frequency regulation) to power system operators.

Water supply systems are suitable candidates to provide flexibility services as they consume large amounts of electricity to power a variety of controllable processes, including water treatment and desalination and water pumping. A study in [2] showed that reducing the energy consumption of a pumping facility by 5% saves around 3–4 MWh of energy annually. In Saudi Arabia, about 9% of the annual energy consumption is due to WSS applications, such as groundwater extraction and water desalination [3]. Provision of flexibility by the water supply system is further motivated by the fact that water demand follows typical seasonal patterns [4]–[7]. It is expected that the capacity of the water supply system is designed to meet the peak water demand. Hence, during seasons where the water demand is off-peak, considerable flexibility can be attained from the water supply system. When the water demand is peaking, flexibility may be very limited.

The shortage of clean water sources is a major problem facing about 40% of the world population. This percentage is expected to increase significantly in the future [8]. The viable solution to this problem in countries that lack natural water resources is mainly through desalinating sea as well as brackish water. These places of water shortage cover significant areas, including the MENA (Middle East and North Africa) region, Asia Pacific, Latin America and the western United States. The International Desalination Association (IDA) reports indicate that the world capacity of desalinated water production exceeds 114 million m^3/day , with estimates that it may approach 200 million m^3/day by 2030 with more than 20,250 plants in 150 countries [8].

The desalination technology started in the 1950's through thermal desalination systems, especially multistage flash (MSF), then multi effect desalination emerged. It was not until the 1990's where Reverse Osmosis (RO) desalination was commercialized in a large scale [9], which consume less energy than thermal desalination. Although the majority of desalination systems in Saudi Arabia (the world's largest desalinated water producer) and the Gulf region are thermal, a strong motion into adopting RO technology is in effect [10]. Consequently, the reliance on thermal power plants (that are the main input source of energy of MSF desalination) to produce electric energy has decreased. New installation of RO desalination plants include the plants in Khafji, Saudi Arabia, which produces 178,560 m^3/day , and in Neom, which is designed to produce 125,000 m^3/day . Both RO plants are solar-powered. Examples at other regions include an RO plant in Chile, which produces 216,000 m^3/day desalinated water [8]. These examples indicate the increasing role RO technology plays in the future of desalination technology.

B. Literature Review

Several research works have studied thermal WDPs as part of solving an economic dispatch (ED) model. More

specifically, ED models are proposed in [11], [12] that optimize the operation of co-generation facilities (e.g., hydroelectric and thermal desalination plants) for serving the water and power demand, considering the constraints of both infrastructures. The models in [11], [12] are expanded to include power and water distribution network limits in [13], but neither account for the energy flexibility that water facilities can introduce to power system operation. In addition, several studies have investigated renewable-powered RO desalination. In [14], a comparative investigation has been presented for several solar technologies that supply off-grid RO desalination plants. In [15], an off-grid RO plant powered by wind and solar PV has been considered. In [16], RES are used to power a RO plant, pump water in a WDS, and store the surplus energy in a pumped hydro storage in order to minimize RES curtailment [17]. However, none of the aforementioned studies explore coordinating the operation of RO plants with power systems.

The technical literature also includes multiple recent frameworks aimed at co-optimizing the operation of power and water systems. In [1], a co-optimization framework is developed that optimizes the energy flexibility of water pumps and storage tanks via solving a network-constrained WDS operation model at a first stage and integrating the WDS energy flexibility in the bulk power system via a network-constrained unit commitment (UC) model at a second stage. To alleviate the computational burden associated with the non-linear constraints of WDSs' components in [1], a linearization technique is developed and validated in [18]. A model for optimal coordinated operation of inter-dependent power and water distribution systems is developed in [19]. To increase the levels of system flexibility in the presence of wind generation, the authors in [20] suggested co-optimizing the wind scheduling with the scheduling of heat pumps and WDS's water pumps. Similarly, a co-optimization model for a wastewater treatment facility is developed in [21] that offers its energy flexibility to the bulk power system via controlling its treatment process. The models in [4], [22], [23] coordinate the operation of power grid with the RO desalination processes. In [4], the unit commitment problem with flexible RO plants is considered, which utilizes a simplified RO desalination model that does not capture the complex operational characteristics of these plants. A more detailed model is presented in [24] that considers hydraulic and osmotic pressures but assuming that the feedwater temperature and the pressure drop are constant throughout the day. In a different approach, models are developed in [25]–[27] that leverage the flexible operation of water storage tanks, pumps, and desalination plants for maximizing the water system operator profits in both demand response programs and frequency regulation markets.

In summary, the models in the literature fail to capture the complex constraints that describe the dynamic operation of RO desalination plants in the coordinated operation of power systems with desalination plants. The simplified models in the literature ignore the fact that the energy needed per unit volume of desalinated water is temperature-dependent. Hence, to produce a given amount of water, different desalination schedules would result in different energy consumption. Other factors, such as hydraulic and osmotic pressure differences, salt

concentration effects, and membrane design characteristics are also oversimplified.

C. Contribution and Paper Structure

This paper proposes a risk-based stochastic co-optimization framework to coordinate the day-ahead (DA) operation of bulk power systems with the operation of RO desalination plants in order to meet the power and water demand at minimum cost. The proposed framework leverages the energy flexibility provided by the RO plants and associated components, such as high-pressure pumps and water storage tanks (WST) in the system. The energy flexibility is integrated in power systems DA operations via a UC model that takes into account the operational constraints of both power and water desalination systems, thus co-optimizing their operation without endangering the reliable supply of services. The proposed risk-based co-optimization framework facilitates higher penetration levels of RES, reduces the operating costs of both power and water supply systems, and shaves the electric load peak. The surplus of RES generation is harvested by storing the extra desalinated water in WSTs, which is used to supply the freshwater demand during high electricity demand period. The main contributions of this paper are summarized below:

- A comprehensive co-optimization framework for the day-ahead operation of bulk power systems and RO water desalination system is proposed, which takes into account the operational constraints of both systems. The RO desalination model considers a wide range of dynamic processes and constraints, such as the dependencies of the different pressures on the operating point of the plant, mass balance constraints for water and salt, salt transport constraints (including their dependency on the concentration polarization factor), and the effect of temperature on the performance of the RO plant.
- For the sake of increasing the computational efficiency for large-scale systems, a linearized formulation for the RO plant model is proposed, which is integrated in the mixed-integer linear programming formulation of the proposed coordinated operation model.
- In order to address the uncertainty associated with the forecast of RES generation and electric loads, the proposed linearized co-optimization model is posed as a two-stage stochastic program. The conditional value at risk (CVaR) is utilized as the risk management measure.

The rest of the paper is organized as follows: overview of the proposed coordinated operation model as well as the proposed RO desalination model are presented in Section II. The linearization of the proposed RO desalination model is presented in Sections III. The proposed risk-based operation coordination model is presented in Section IV. The simulation setup and simulation results are presented in Sections V and VI and the conclusions are drawn in Section VII.

II. THE MODEL OVERVIEW

The structure of the proposed risk-based stochastic model for coordinated operation of power systems and RO plants is shown in Fig. 1. In Fig. 1, the power system operator, which is

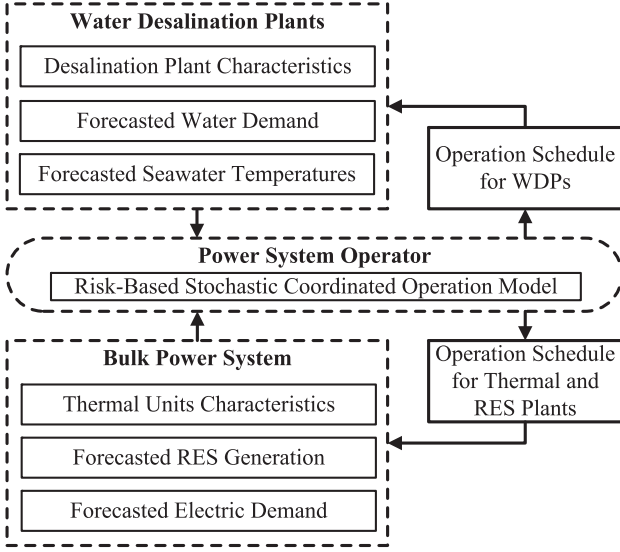


Fig. 1. Proposed framework for coordinated operation of power systems and water desalination plants.

assumed to be independent, schedules the DA joint operation of the power system and water desalination plants. The power system operator utilizes the parameters of the thermal generating units, such as operating and start-up cost functions, power output limits, ramp rates, minimum up- and down-times, transmission power flow constraints and line flow limits, as well as the DA forecasts of the RES power output and the uncontrollable electric demand. Likewise, the WDP operator submits to the power system operator the characteristics of the WDPs, such as the water storage tank capacities, available stored water volumes, start-up costs, as well as the DA forecasts for water demands and seawater temperatures. Using the power system and the WDP plants characteristics, the power system operator runs a the proposed risk-based stochastic coordinated operation model that aims to minimize the total operating costs for meeting the demands of the power and water systems. In the proposed model, the WPDs are assumed to be the only source of demand flexibility, while the rest of the power loads are considered uncontrollable. The result of the proposed model provides the DA schedules of the power plants and the desalination plants.

A. The RO Plant Model

The proposed DA co-optimization model incorporates a detailed model of the RO desalination plants, as shown in Fig. 2. In Fig. 2, the RO plants uses a semi-permeable membrane that permits water molecules to pass through it while inhibiting the passage of dissolved salt molecules that are larger than the water molecules. Consequently, water before the membrane, or feedwater, has higher salinity than that of the water after the membrane, or permeate water. If feedwater is left un-pressurized, the osmosis pressure would force permeate water to flow back through the membrane to the feedwater side. Hence, the feedwater must be pressurized using electric pumps to force water to flow in one direction (from left to right). This water pumping consumes electric power for increasing the head

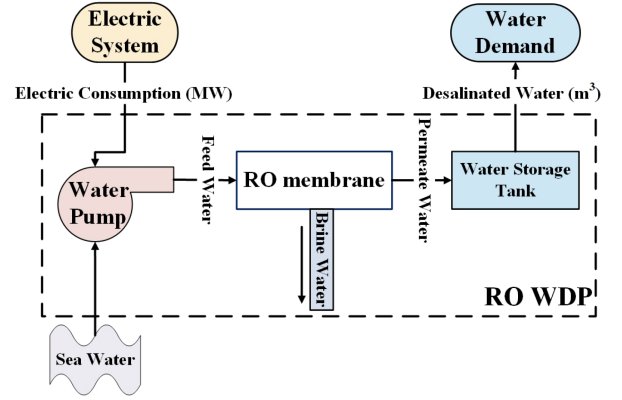


Fig. 2. The RO desalination system connected to the grid for power source.

pressure along the RO membranes. Hence, for each RO plants, the input commodity is electric power and the output commodity is fresh water. The desalinated water is stored in WSTs to serve freshwater demand throughout the day. Note that the RO process is defined in terms of the hydraulic and osmotic pressures, water and salt transport coefficients, salt passage and rejection, and water permeate recovery.

The proposed RO plant model assumes that: a) the RO plant is in steady state conditions with varying water flows and pressures; b) the permeate compartment is completely mixed; and c) the salt ions concentration is linearly varying within the feed water compartment along the RO membrane area.

B. Formulation of the RO Plant Model

The formulation of the proposed operation model for RO plants is presented here. For each RO plant, the electric demand is presented in (1). This nonlinear relation represents the required electric demand to operate high pressure pumps in each RO plant k , which depend on the feedwater flow $Q_{z,k,t}^{w,f}$ and the applied hydraulic pressure $H_{z,k,t}^{w,f}$.

$$L_{z,k,t}^w = \left(Q_{z,k,t}^{w,f} H_{z,k,t}^{w,f} \right) / \left(36 * 10^5 \eta_{z,k}^{w,Pump} \right). \quad (1)$$

The hydraulic pressure, $H_{z,k,t}^w$, depends on the rate of recovery and the feedwater quality. The increase in the rate of recovery, which represents the ratio of the permeate water to the feedwater, results in an increase in salt concentration of the rejected brine water. The passage of salt is defined as the opposite of salt rejection, which describes the percentage of feedwater ions prevented to cross the RO membrane. A higher concentration of salt in the feed-concentrated solution raises the membrane osmotic pressure, π^w , which decreases the difference $(\Delta \bar{H}_{z,k,t}^w - \Delta \bar{\pi}_{z,k,t}^w)$ and, consequently, decreases the production flow rate of water [9]. The permeate flow rate ($Q_{z,k,t}^{w,p}$) is formulated as follows:

$$Q_{z,k,t}^{w,p} = K_{z,k}^{w,wat} \cdot S_{z,k}^w \cdot T_{z,t}^{F,w} \cdot N_{z,k}^w \cdot (\Delta \bar{H}_{z,k,t}^w - \Delta \bar{\pi}_{z,k,t}^w), \quad (2)$$

where $Q_{z,k,t}^{w,p}$ depends on four factors based on the membrane characteristics and water temperature: the water membrane

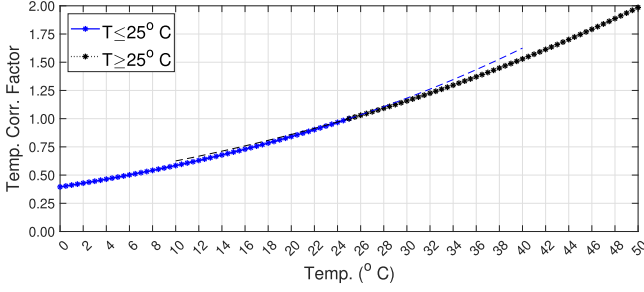


Fig. 3. Temperature correction factor Versus Temperature.

permeability coefficient, $K_{z,k}^{w,wat}$, the surface area of the RO membrane, S , the number of RO membrane modules, N , and the temperature correction factor, T^F . In this paper, the temperature correction factor is defined by (3), which is obtained from a membrane manufacturer [28]. Fig. 3 shows T^F versus temperature for a wide range of temperatures.

$$T_{z,t}^{F,w} = \begin{cases} \exp \left[\left(\frac{1}{298} - \frac{1}{273+T_{z,t}^w} \right) \alpha_1^w \right], & \text{if } T_{z,t}^w \leq 25^\circ\text{C} \\ \exp \left[\left(\frac{1}{298} - \frac{1}{273+T_{z,t}^w} \right) \alpha_2^w \right], & \text{if } T_{z,t}^w \geq 25^\circ\text{C} \end{cases} \quad (3)$$

It is worth mentioning that membrane manufacturers, e.g., [28], indicate that membrane activity is sensitive to feedwater temperature changes. Water flux increases with temperature almost linearly due to the increase in the rate of diffusion through the membrane. That is, higher temperatures have a favorable effect on the system performance and efficiency. However, it is not recommended to operate at temperatures higher than 100°F (37.8°C). On the other hand, higher temperatures and increased diffusion rates through the RO membranes may be accompanied by a decrease in salt rejection, which affects the quality of permeated water. This finding is confirmed in [29], where it is also stated that membrane capacity increases by about 3%/°C increase in water temperature. Therefore, the performance of the RO system to changes in temperature as well as variation in water quality makes the use of a simple conversion factor to represent the energy needed for water desalination system an oversimplification that does not reflect the nature of the process and its associated parameters.

The operation of RO plants is subject to the water balance constraint (4), where the total flow rate of the stored desalinated water in the tanks equals to the total water demand of the WDS. Note that WDPs are assumed to be the only source of freshwater in this work. Constraint (5) ensures that the water demand is served at all time.

$$\sum_{z,k} Q_{z,k,t}^{w,WST} = D_{z,t}^w \sum_{z,k} \gamma_{z,k,t}^w \quad (4)$$

$$\sum_{z,k} \gamma_{z,k,t}^w = 1. \quad (5)$$

1) *The Applied Hydraulic Pressure Constraints:* The hydraulic pressure difference in (6) is defined as the difference between the high-pressure side (average of feed, $H_{z,k,t}^{w,f}$, and brine, $H_{z,k,t}^{w,b}$, pressures), and the permeate pressure, $H_{z,k,t}^{w,p}$, which is equal to the atmospheric pressure. These pressures are limited from the top according to (7), which are based on the

membrane characteristics [9], where $\sigma_{z,k}^w$ is a parameter equal to the difference percentages between H_b^w and H_f^w .

$$\Delta \bar{H}_{z,k,t}^w = \left[\left(H_{z,k,t}^{w,f} + H_{z,k,t}^{w,b} \right) / 2 \right] - H_{z,k,t}^{w,p}, \quad (6)$$

$$H_{z,k,t}^{w,b} / \sigma_{z,k}^w \leq H_{z,k,t}^{w,f} \leq \bar{H}_{z,k,t}^{w,f} u_{z,k,t}^w. \quad (7)$$

2) *The Mass and Concentration Balance Constraints:* The mass balance constraint for the RO membrane system states that the system mass must be preserved over time, as given by (8). Similarly, the mass balance for the salt is given by (9).

$$Q_{z,k,t}^{w,f} = Q_{z,k,t}^{w,b} + Q_{z,k,t}^{w,p}, \quad (8)$$

$$Q_{z,k,t}^{w,f} X_{z,k,t}^{w,f} = Q_{z,k,t}^{w,b} X_{z,k,t}^{w,b} + Q_{z,k,t}^{w,p} X_{z,k,t}^{w,p}. \quad (9)$$

3) *The Applied Osmotic Pressure and Salt Transport Constraints:* The average difference of the osmotic pressure applied on the RO membrane is formulated as a difference between permeate osmotic pressure, $\pi_{z,k,t}^{w,p}$, and an average value between the feed, $\pi_{z,k,t}^{w,f}$, and brine, $\pi_{z,k,t}^{w,b}$, osmotic pressures as in (10), where $\pi_{z,k,t}^{w,f}$, $\pi_{z,k,t}^{w,b}$, and $\pi_{z,k,t}^{w,p}$ are proportional to the corresponding salinities with κ^w to be the proportionality constant. The value of this constant is based on the assumption that the osmotic pressure π^w is 75.84 kPa for a 1000 ppm of Total Dissolved Solids (TDS) [9]. These concepts are formulated in (11). Note that the feedwater salinity is assumed to remain constant for each unit k during the optimization time horizon. The salinities of brine and permeate water are constrained by minimum and maximum limits as shown in (12)–(13).

$$\Delta \bar{\pi}_{z,k,t}^w = \left[\left(\pi_{z,k,t}^{w,f} + \pi_{z,k,t}^{w,b} \right) / 2 \right] - \pi_{z,k,t}^{w,p} C_{z,k}^{F,w}, \quad (10)$$

$$\pi_{z,k,t}^{w,f} = \kappa^w X_{z,k,t}^{w,f}, \pi_{z,k,t}^{w,b} = \kappa^w X_{z,k,t}^{w,b}, \pi_{z,k,t}^{w,p} = \kappa^w X_{z,k,t}^{w,p}, \quad (11)$$

$$\bar{X}_{z,k,t}^{w,p} u_{z,k,t}^w \leq X_{z,k,t}^{w,p} \leq \bar{X}_{z,k,t}^{w,p} u_{z,k,t}^w, \quad (12)$$

$$X_{z,k,t}^{w,b} \geq X_{z,k,t}^{w,f} u_{z,k,t}^w. \quad (13)$$

Similar to (2), the salt transport across RO membrane is defined in (14) as the difference between average dissolved solids concentration ($\bar{X}_{z,k,t}^w$) and permeate water salinity ($X_{z,k,t}^{w,p}$). While the ($\bar{X}_{z,k,t}^w$) is defined in (15), and $C_{z,k}^{F,w}$ is the concentration polarization factor. When the feedwater flows across the RO membrane and its salts is rejected, a boundary layer is created near the surface of the membrane in a high concentration that exceeds the salt concentration in feedwater side. This phenomena is called concentration polarization, which is expressed in terms of a concentration polarization factor ($C_{z,k}^{F,w}$) formulated for each RO plant k as in (16). The effect of this phenomena is reducing the actual production of desalinated water and salt rejection compared to the theoretical estimates [28].

$$M_{z,k,t}^{w,S} = K_{z,k}^{w,s} \cdot S_{z,k}^{w,s} \cdot T_{z,k,t}^{F,w} \cdot N_{z,k}^w \cdot C_{z,k}^{F,w} (\bar{X}_{z,k,t}^w - X_{z,k,t}^{w,p}), \quad (14)$$

$$\bar{X}_{z,k,t}^w = \left(Q_{z,k,t}^{w,f} X_{z,k,t}^{w,f} + Q_{z,k,t}^{w,b} X_{z,k,t}^{w,b} \right) / \left(Q_{z,k,t}^{w,f} + Q_{z,k,t}^{w,b} \right), \quad (15)$$

$$C_{z,k}^{F,w} = \exp \left[0.7 * Q_{z,k,t}^{w,p} / Q_{z,k,t}^{w,f} \right]. \quad (16)$$

4) *Additional Operational Constraints of the RO Plants:* Constraint (17) ensures that the production of water from a WDP is limited by its minimum and maximum production capacity. In addition, constraint (18) enforces the WDP ramping up/down limits, which are dictated by the membrane characteristics. Logical constraints (19) and (20) ensure that a WDP cannot be turned on and off simultaneously. They also define the start-up and shut-down binary variables as functions of transitions in the plant status.

$$u_{z,k,t}^{w,p} \bar{Q}_{z,k}^{w,p} \leq Q_{z,k,t}^{w,p} \leq u_{z,k,t}^w \bar{Q}_{z,k}^{w,p}, \quad (17)$$

$$-RD_{z,k}^w \leq Q_{z,k,t}^{w,p} - Q_{z,k,t-1}^{w,p} \leq RU_{z,k}^w, \quad (18)$$

$$SU_{z,k,t}^w - SD_{z,k,t}^w = u_{z,k,t}^w - u_{z,k,t-1}^w, \quad (19)$$

$$SU_{z,k,t}^w + SD_{z,k,t}^w \leq 1. \quad (20)$$

5) *Water Storage Tank Constraints:* The water flowing out of a WST k at time t is limited by the available water volume in the tank in (21), while the water volume in the tank is confined by its maximum and minimum capacity volume limits in (22). The inter-temporal state of charge constraint of the water storage tank is given in (23). Relation (24) sets the initial and final WSTs water volumes.

$$0 \leq Q_{z,k,t}^{w,ST} \Delta t \leq V_{z,k,t}^{w,ST}, \quad (21)$$

$$\mu^{w,\min} V_{z,k}^w \leq V_{z,k,t}^{w,ST} \leq \mu^{w,\max} V_{z,k}^w, \quad (22)$$

$$V_{z,k,t}^{w,ST} = V_{z,k,t-1}^{w,ST} + (Q_{z,k,t}^{w,p} - Q_{z,k,t}^{w,ST}) \Delta t, \quad (23)$$

$$V_{z,k,1}^{w,ST} = V_{z,k,24}^{w,ST} = V_{o,k}^w = \tau_{z,k}^w V_{z,k}^w. \quad (24)$$

III. LINEARIZATION OF RO DESALINATION MODEL

The proposed RO plant operation model (1)–(24) is nonlinear due to the non-linearities introduced by the energy consumption of RO pumping system (1), the concentration balance (9), the average TDS (15), and the CPF constraints (16). If this model is used in the coordinated operation model, it will give rise to a mixed-integer nonlinear program, which may be computationally intensive and may not guarantee converging to a feasible global optimal solution at all circumstances. Therefore, a piece-wise linear technique [26] is applied to the presented nonlinear relations, rendering a mixed-integer linear programming (MILP) model.

A. Linearization of RO Energy Consumption Function

In this subsection, a linearization model is developed for the energy consumed by the high pressure pumps systems of the RO plants given in (1). In order to develop a piece-wise linear model for the bivariate function $L^w(H_{z,k,t}^{w,f}, Q_{z,k,t}^{w,f})$, the axes $H_{z,k,t}^{w,f}$ and $Q_{z,k,t}^{w,f}$ are divided into $(n-1)$ and $(m-1)$ segments, respectively, where $\psi_{z,k,1}^w, \psi_{z,k,2}^w, \dots, \psi_{z,k,n}^w$ are breakpoints along the $H_{z,k,t}^{w,f}$ axis, and $\omega_{z,k,1}^w, \omega_{z,k,2}^w, \dots, \omega_{z,k,m}^w$ are breakpoints along the $Q_{z,k,t}^{w,f}$ axis, as shown in Fig. 4. The values of $\psi_{z,k,n}^w$ and $\psi_{z,k,1}^w$ coincide with the upper and lower bounds of $H_{z,k,t}^{w,f}$, while

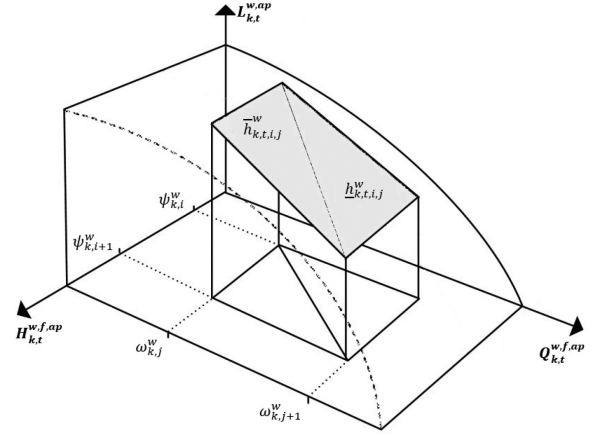


Fig. 4. Geometric representation of triangle method that applied to RO energy consumption function (the drawing adopted from [18]).

$\omega_{z,k,m}^w$ and $\omega_{z,k,1}^w$ coincide with the upper and lower bounds of $Q_{z,k,t}^{w,f}$. Hence, the bivariate function $L^w(H_{z,k,t}^{w,f}, Q_{z,k,t}^{w,f})$ defined in (1) is divided into several rectangles. Now, consider the rectangle whose heads are $(\psi_{z,k,i}^w, \omega_{z,k,j}^w)$, $(\psi_{z,k,i+1}^w, \omega_{z,k,j}^w)$, $(\psi_{z,k,i+1}^w, \omega_{z,k,j+1}^w)$, and $(\psi_{z,k,i}^w, \omega_{z,k,j+1}^w)$. This rectangle can be divided into two triangles partitioned by the diagonal line $(\psi_{z,k,i}^w, \omega_{z,k,j}^w)$, $(\psi_{z,k,i+1}^w, \omega_{z,k,j+1}^w)$. Consequently, two binary variables $\bar{h}_{z,k,t,i,j}^w$ ($i = 1, 2, \dots, n; j = 1, 2, \dots, m$) and $\bar{h}_{z,k,t,i,j}^w$ ($i = 1, 2, \dots, n; j = 1, 2, \dots, m$) are introduced, and they are associated with the lower-right and the upper-left triangles, respectively, and a continuous variable $\beta_{z,k,t,i,j}^w \in [0, 1]$ associated with each head (i, j) is defined. Accordingly, relations (25) and (26) represent all operating points given in the range $\psi_{z,k,i}^w \leq H_{z,k,t}^{w,f} \leq \psi_{z,k,i+1}^w$ and $\omega_{z,k,j}^w \leq Q_{z,k,t}^{w,f} \leq \omega_{z,k,j+1}^w$. Each operating point is defined in a unique linear combination of organized heads that are scaled by their associated variables $\beta_{z,k,t,i,j}^w$. Then, the bivariate function $L^w(H_{z,k,t}^{w,f}, Q_{z,k,t}^{w,f})$ can be piece-wisely approximated by $L_{z,k,t}^{w,ap}$ in a linear combination of evaluated values of L^w at the triangle's heads that restrain the operating point $(H_{z,k,t}^{w,f}, Q_{z,k,t}^{w,f})$ and weighted by $\beta_{z,k,t,i,j}^w$, as in (27). The constraints (28)–(31) ensure that the optimizer chooses only one triangle identified by three triangle heads and associated with three weight values $\beta_{z,k,t,i,j}^w$.

$$H_{z,k,t}^{w,f,ap} = \sum_{i=1}^n \sum_{j=1}^m \beta_{z,k,t,i,j}^w \psi_{z,k,i}^w, \quad (25)$$

$$Q_{z,k,t}^{w,f,ap} = \sum_{i=1}^n \sum_{j=1}^m \beta_{z,k,t,i,j}^w \omega_{z,k,j}^w, \quad (26)$$

$$L_{z,k,t}^{w,ap} = \sum_{i=1}^n \sum_{j=1}^m \beta_{z,k,t,i,j}^w L^w(\psi_{z,k,i}^w, \omega_{z,k,j}^w), \quad (27)$$

$$\sum_{i=1}^n \sum_{j=1}^m (\bar{h}_{z,k,t,i,j}^w + \bar{h}_{z,k,t,i,j}^w) = 1, \quad (28)$$

$$0 \leq \beta_{z,k,t,i,j}^w \leq 1, \quad (29)$$

$$\sum_{i=1}^n \sum_{j=1}^m \beta_{z,k,t,i,j}^w = 1, \quad (30)$$

$$\begin{aligned} \beta_{z,k,t,i,j}^w \leq & \bar{h}_{z,k,t,i,j}^w + \bar{h}_{z,k,t,i,j}^w + \bar{h}_{z,k,t,i,j-1}^w \\ & + \bar{h}_{z,k,t,i-1,j}^w + \bar{h}_{z,k,t,i-1,j-1}^w + \bar{h}_{z,k,t,i-1,j-1}^w. \end{aligned} \quad (31)$$

B. Linearization of Concentration Balance, Average TDS, and CPF Constraints

The concentration balance constraint (9) is a nonlinear constraint due to the multiplications of two variables, as in $Q_{z,k,t}^{w,b} X_{z,k,t}^{w,b}$ and $Q_{z,k,t}^{w,p} X_{z,k,t}^{w,p}$. Fortunately, it can be linearized by simple mathematical compensations as follows. Constraints (8) and (32) can be substituted by (9). The resulted equation is then manipulated and rearranged to produce relation (33). Since the feed water salinity $X_{z,k,t}^{w,f}$ is assumed to be known by the operator of each RO plant and $Re_{z,k,t}^w$ is a variable that changes slowly during one-day operation, and is assumed to remain constant throughout the considered time horizon, then constraint (33) is a linear version that can fully represent the nonlinear constraint (9). Similarly, the constraint of average TDS concentration (15) can be manipulated and rearranged to be represented in a linear form as shown in (34). Constraint (16) can be represented as in (35).

$$Q_p^w(z, k, t) = Q_f^w(z, k, t) Re^w(z, k, t), \quad (32)$$

$$X_{z,k,t}^{w,f} = (1 - Re_{z,k,t}^w) X_{z,k,t}^{w,b} + Re_{z,k,t}^w X_{z,k,t}^{w,p}, \quad (33)$$

$$\bar{X}_{z,k,t}^w = \frac{X_{z,k,t}^{w,f} + (1 - Re_{z,k,t}^w) X_{z,k,t}^{w,b}}{2 - Re_{z,k,t}^w}, \quad (34)$$

$$C_{z,k}^{F,w} = \exp [0.7 * Re_{z,k}^w]. \quad (35)$$

The proposed linear model is approximating the non-linear RO plant model. Substituting the previously mentioned convex combinations and linear approximation constraints (25)–(35), with their nonlinear ones (1), (9), (15), and (16) converts the nonlinear proposed model for coordinating RO plant operation with the power system operation model into a linear one.

IV. THE PROPOSED RISK-BASED STOCHASTIC COORDINATED OPERATION MODEL

This Section discusses the uncoordinated optimization of power and water systems and the proposed stochastic co-optimization of both systems.

A. Uncoordinated Water and Power Optimization Models

In this mode, the WDP operator optimizes the operation of the water desalination system independently of the power system. The operation model of RO plants presented in Section II-A would allow the WDP operator to locally minimize its energy consumption while supplying the hourly water demand. The uncoordinated energy consumption of WDP is determined using the following operational model:

$$\begin{aligned} & \min \sum_{z,k,t} L_{z,k,t}^w \\ \text{s.t. } & (1), \quad \text{WDP Power consumption,} \\ & (2)–(3), \quad \text{Permeate water flow,} \\ & (4)–(5), (21)–(24), \quad \text{WST,} \\ & (6)–(7), \quad \text{Hydraulic pressure,} \\ & (8)–(9), (12)–(16), \quad \text{Mass balances \& salt transport,} \\ & (10)–(11), \quad \text{Osmotic pressure,} \\ & (25)–(35), \quad \text{WDP Linearization.} \end{aligned}$$

The optimal hourly power consumption obtained from this optimization is shown by $L_{z,k,t}^{w,\circ} = L_{z,k,t}^w$. This value will be sent to the power system operator as uncontrolled demands. Consequently, the power system operator will add the WDP power demand to its forecasted load profile, which will be used in a standard cost-minimizing UC optimization model to obtain the thermal generation schedules.

B. Risk-Based Stochastic Coordinated Operation Model

In this section, the risk-based stochastic coordinated operation model is formulated to integrate the operation of the flexible energy consumption of WDP into the power system operation. In this case, the WDP operators send their characteristics to power system operator, who runs the proposed risk-based stochastic coordinated operation model in order to obtain the coordinated schedule of both the WDPs and thermal generation units.

The proposed risk-based stochastic coordinated operation model is formulated as a mixed-integer linear two-stage stochastic optimization problem with recourse [30]. In the proposed model, the uncertain parameters are the RES power outputs and the electric loads. Multiple scenarios for each uncertain parameter are generated based on historical data. A scenario tree is then built by coupling the scenarios of the problem's uncertain parameters. The here-and-now decisions include the thermal UC status and the WDP schedules including the desalinated water, pressures, water flow, and their related water salinities. The wait-and-see decisions include the actual power outputs and spinning reserve capacities of thermal units and scheduled and curtailed RES generation. The proposed model integrates CVaR as the risk measure, which has been used in the power system operation literature in other contexts, e.g., [31]–[33].

1) *Objective Function*: The objective of the proposed EWN model is to minimize the overall expected operating cost for meeting the electricity and water demands of the coupled power and water desalination system over the scheduling horizon. To capture the uncertainty of DA forecasted parameters, a scenario-based SP has been formulated and each scenario is associated with its probability, as given in (36) and (37).

$$\text{minimize} \quad \sum_s^{N^s} \Omega_s F_s (1 - \varphi) + \varphi CVaR_\varrho, \quad (36)$$

$$\begin{aligned} F_s = & \sum_{z,g,t} P_{s,z,g,t}^{th} b_{z,g,t}^{th} + \sum_{z,k,t} C_{z,k}^w RL_{z,k,t}^w \\ & + \sum_{z,pv,t} P_{s,z,pv,t}^v b_{pv,t}^v + \sum_{z,wi,t} P_{s,z,wi,t}^d b_{wi,t}^d \\ & + \sum_{z,g,t} (C_{z,g}^{th,SU} SU_{z,g,t}^{th} + C_{z,g}^{th,SD} SD_{z,g,t}^{th}). \end{aligned} \quad (37)$$

In (37), the first and second lines represent the operating costs for the thermal generation units, the cost for energy shifting of RO plants, and the RES generation cost. The third line represents the thermal generation's startup and shutdown costs. The "here-and-now" decision variables here are $(u_{z,g,t}^{th})$, $(SU_{z,g,t}^{th})$, and $(SD_{z,g,t}^{th})$. The "wait-and-see" decision variables here are $(P_{s,z,g,t}^{th})$, $(P_{s,z,pv,t}^v)$, and $(P_{s,z,wi,t}^d)$.

Due to the uncertainties involved in this problem, the conditional value at risk (CVaR) mechanism is used. At a specific level of confidence $\varrho \in [0, 1]$, the $CVaR_\varrho$ obtains the expected costs for the $(1 - \varrho) * 100\%$ most costly scenarios and can be introduced to the SP model in a linear form (in this work,

$\varrho = 0.89$). To consider risk control, the second term of (36) is included. In addition, constraints (38)–(39) are included. The parameter $\varphi \in [0, 1]$ is a predefined parameter to weight the risk aversion behavior of the power system operator. The risk behavior can be defined by sitting $\varphi = 0$ for a risk-neutral behavior and $\varphi > 0$ for a risk-aversion behavior.

$$CVaR_{\varrho} = VaR + (1 - \varrho)^{-1} \sum_s^{N^s} \Omega_s \rho_s, \quad (38)$$

$$F_s - VaR - \rho_s \leq 0. \quad (39)$$

2) *Energy Flexibility Model of RO Plants*: This optimization is subject to power balance constraint given in (40), which states that, at each bus z , the expected power outputs of thermal, solar PV, and wind power plants should balance out the sum of the expected non-controllable electric loads, WDPs' flexible loads, and line flows out of bus z . Notice that the expected electric consumption of RO plants, $L_{z,k,t}^w$, is a “here-and-now” decision variable.

$$\sum_g P_{s,z,g,t}^{th} + \sum_v P_{s,z,pv,t}^v + \sum_{wi} P_{s,z,wi,t}^d - L_{s,z,t}^D - \sum_k L_{z,k,t}^w = \sum_{z'} P_{s,zz',t}^L. \quad (40)$$

In the energy flexibility model, it is assumed that the WDPs can shift some of their energy consumption upon the request of the power system operator. This is helped by the fact that each WDP is equipped with a WST. If a WDP is requested to shift its energy consumption from time t to another time slot t' , we will denote the reduced energy by $RL_{z,k,t}^w$ and the added energy as $AL_{z,k,t'}^w$. It should be noted that, in general, $AL_{z,k,t'}^w$ is not equal to $RL_{z,k,t}^w$. This is because the energy needed to produce 1 m^3 of desalinated water is time-dependent. This is because it is a function of temperature, among other factors, as explained in Section III-A. Therefore, most existing DR models proposed in the literature for generic demands, which typically assume that the reduced energy to be equal to the added energy, can not be used in this application.

The proposed energy flexibility model of WDP operation is represented in (41)–(43). In (41), $L_{z,k,t}^w$ refers to the total consumed power of WDP k at time t , which is equal to the consumed power that is uncoordinated, $L_{z,k,t}^{w,\circ}$, minus the reduced ($RL_{z,k,t}^w$) plus the added ($AL_{z,k,t}^w$) energy capacities that are utilized by the power system operator. The WDP operator is compensated for the energy flexibility services based on the reduced energy capacity. The difference between the coordinated and uncoordinated energy consumption of WDP bounds the energy flexibility variables ($RL_{z,k,t}^w$ and $AL_{z,k,t}^w$) as in (42)–(43). The difference ($L_{z,k,t}^{w,\text{coord}} - L_{z,k,t}^{w,\circ}$) is considered as the value of $RL_{z,k,t}^w$ when it is negative; otherwise, $RL_{z,k,t}^w$ is zero. Similarly, this difference is considered as the value of $AL_{z,k,t}^w$ when it is positive; otherwise, $AL_{z,k,t}^w$ is zero. The energy reduction $RL_{z,k,t}^w$ indicates that the WDP operator has an opportunity to reduce the energy consumption and serve the water demand from the water stored in WSTs. On the contrary, the added energy $AL_{z,k,t}^w$ indicates that the WDP energy consumption can be increased to desalinate more water and store the surplus in WSTs for serving future water demand.

$$L_{z,k,t}^{w,\text{coord}} = L_{z,k,t}^{w,\circ} + RL_{z,k,t}^w + AL_{z,k,t}^w, \quad (41)$$

$$RL_{z,k,t}^w = \min [L_{z,k,t}^{w,\text{coord}} - L_{z,k,t}^{w,\circ}, 0], \quad (42)$$

$$AL_{z,k,t}^w = \max [L_{z,k,t}^{w,\text{coord}} - L_{z,k,t}^{w,\circ}, 0]. \quad (43)$$

3) *Power System Constraints*: The DC active power flow constraints are defined in (44)–(45). The outputs of the thermal generating units are bounded by their minimum and maximum capacity limits in (46). The ramping up and down constraints are given in (47) and (48), respectively. The required spinning reserve constraint is limited to be at least within a reasonable percentage of the un-controllable electric load and less than the ramping limits for each generator as in (49) to (51). Constraints (52)–(53) define the bounds for the scheduled RES generation while (54)–(55) define the RES curtailments. Constraints (56) and (57) ensure that the unit will not start-up and shut-down simultaneously [34]. The minimum-up time constraint of the thermal generation units is formulated similar to [31] and [35]. The “here-and-now” decision variable here is $u_{z,k,t}^{th}$. The newly mentioned “wait-and-see” decision variables here are $P_{s,z,t}^L$, $\theta_{z,t}$, $SR_{s,z,g,t}^{th}$, $\hat{P}_{s,z,pv,t}^v$, and $\hat{P}_{s,z,wi,t}^d$.

$$P_{s,zz',t}^L = (\theta_{s,z,t} - \theta_{s,z',t}) / x_{zz'}, \quad (44)$$

$$-\bar{P}_{zz'}^L \leq P_{s,zz',t}^L \leq \bar{P}_{zz'}^L, \quad (45)$$

$$\bar{P}_{z,g}^{th} u_{z,g,t}^{th} \leq P_{s,z,g,t}^{th} \leq \bar{P}_{z,g}^{th} u_{z,g,t}^{th}, \quad (46)$$

$$P_{s,z,g,t}^{th} - P_{s,z,g,t-1}^{th} \leq RU_{z,g}^{th}, \quad (47)$$

$$P_{s,z,g,t-1}^{th} - P_{s,z,g,t}^{th} \leq RD_{z,g}^{th}, \quad (48)$$

$$SR_{s,z,g,t}^{th} = \bar{P}_{z,g}^{th} u_{z,g,t}^{th} - P_{s,z,g,t}^{th}, \quad (49)$$

$$\sum_{z,g} SR_{s,z,g,t}^{th} \geq \lambda^{th} L_{s,z,t}^D, \quad (50)$$

$$SR_{s,z,g,t}^{th} \leq RU_{z,g}^{th} u_{z,g,t}^{th}, \quad (51)$$

$$0 \leq P_{s,z,pv,t}^v \leq \hat{P}_{s,z,pv,t}^v \bar{P}_{z,pv}^v, \quad (52)$$

$$0 \leq P_{s,z,wi,t}^d \leq \hat{P}_{s,z,wi,t}^d \bar{P}_{z,wi}^d, \quad (53)$$

$$\hat{P}_{s,z,pv,t}^v = \tilde{P}_{s,z,pv,t}^v \bar{P}_{z,pv}^v - P_{s,z,pv,t}^v, \quad (54)$$

$$\hat{P}_{s,z,wi,t}^d = \tilde{P}_{s,z,wi,t}^d \bar{P}_{z,wi}^d - P_{s,z,wi,t}^d, \quad (55)$$

$$SU_{z,g,t}^{th} - SD_{z,g,t}^{th} = u_{z,g,t}^{th} - u_{z,g,t-1}^{th}, \quad (56)$$

$$SU_{z,g,t}^{th} + SD_{z,g,t}^{th} \leq 1. \quad (57)$$

V. SIMULATION SETUP

This section presents the simulation setup for two studies that are carried out to assess the efficacy of the proposed energy-water stochastic co-optimization approach is presented.

A. Simulation Studies

1) *Study 1. Deterministic Operation Model*: In this study, four different cases are examined for the deterministic operation model. These cases differ in the way the optimization is carried out and the way the WDP participation factors, γ^w , are determined, as shown in Table I.

TABLE I
CASE STUDIES

Case	Opt. method	WDP Participation Factor
0	Sequential	Pre-determined
1	Simultaneous	Pre-determined
2	Sequential	Decision variable
3	Simultaneous	Decision variable

Case 0: The WDP participation factors are assumed to be pre-determined. In addition, the WDP scheduling optimization and power system scheduling optimization are carried out sequentially. That is, the WDP operator optimizes the desalination schedules and shares these schedules with the power system operator. Then, the power system operator runs the power system scheduling, i.e., conventional UC, considering the WDPs as uncontrollable demands.

Case 1: The WDP participation factors are also pre-determined. However, WDP scheduling and power system scheduling are co-optimized by the power system operator with the objective of minimizing the overall operating costs of EWN system.

Case 2: The WDP participation factors are considered as decision variables. The scheduling optimization of the water and power infrastructures is performed sequentially.

Case 3: The decision variables of the WDP participation factors and EWN system are co-optimized.

Note that Cases 0 and 1 assume that the desalinated water pipelines are under-developed. Hence, each WDP is serving its local water demand. This condition considers $\gamma_{\circ}^w$ as a pre-defined value by the WDP operator ($\gamma_{z,k,t}^w = \gamma_{\circ,k}^w$). In Cases 2 and 3, the water pipeline network is assumed to be well-established, and desalinated water can be readily transported from one area to another. Hence, water desalination scheduling is not dependent of the WDP location. Consequently, $\gamma_{z,k,t}^w$ in (4)–(5) are not pre-defined. Instead, they are decision variables identified by the WDP operator in Case 2 or the power system operator in Case 3.

The sequential optimization (Cases 0 and 2) is conducted as follows. The WDP operation is first optimized by the WDP operator to minimize the WDPs' overall power consumption as presented in Subsection (IV-A). The WDP operator, then, submits its electric load schedule to the power system operator that runs a UC model to minimize the power system's operating cost:

$$\begin{aligned} \min \quad & (37), \\ \text{s.t.} \quad & (40), (44)–(57). \end{aligned}$$

In contrast to Cases 0 and 2, Cases 1 and 3 illustrate the benefits that can be provided by co-optimizing the schedules of the water and power systems. The underlying assumption here is that the power system and WDP are both operated by the power system operator. As such, the operation of these two systems can be centrally co-optimized as follows:

$$\begin{aligned} \min \quad & (37), \\ \text{s.t.} \quad & (1)–(17), (40)–(57). \end{aligned}$$

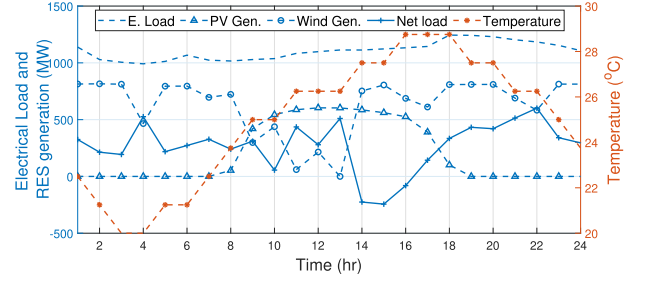


Fig. 5. RES generation, electrical load, net load, and temperature.

2) *Study 2: Stochastic Operation Model:* For the stochastic programming model, Case 4 is investigated to evaluate the coordination benefits in the presence of uncertainties for the forecasted parameters. Similar to the deterministic operation model, it is assumed here that all the RES belong to the power system operator. Note that Case 4 represents the stochastic formulation corresponding to Case 3 of the deterministic model. Case 4 is represented as follows:

$$\begin{aligned} \min \quad & (36), \\ \text{s.t.} \quad & (1)–(57). \end{aligned}$$

B. Test System

The proposed stochastic co-optimization framework is implemented on a modified version of the IEEE 24-bus RTS system that powers five WDPs [36]. Wind and solar generation resources are also included to cover a portion of the coupled system demand. The characteristics of thermal generating units and grid branches are given in [36]. The electric load profiles represent a scaled down version of the Saudi load profile in 2015. The scaling factor used is 24. The RES generation profiles are obtained based on real wind speed and solar radiation data in the Northern Province of Saudi Arabia. The locations for WDPs 1–5 are buses 18, 15, 13, 10, and 9 respectively. The RES are located based on the recommendations given in [36]. Based on the available data, the simulations are carried out for a sample day. The load, RES, net load, and water temperature data are shown in Fig. 5. The characteristics of WDP are presented in [37].

For the stochastic cases (Cases 4 and 5), the forecast uncertainties for wind, PV, and electric load profiles are considered. For each profile, 5000 scenarios are generated based on the statistical characteristics obtained from historical data. Then, to maintain the optimization problem tractable, these scenarios are reduced using the K-means clustering method. For each parameter, the clustering method resulted in three scenarios. Hence, a scenario tree of 27 scenarios is considered.

VI. RESULTS AND DISCUSSION

A. Study 1: Deterministic Operation Model

1) *WDPs' Scheduling Flexibility:* The WDPs scheduled electric load profiles for the 24-hour time horizon for the four cases during a sample day are illustrated in Fig. 6. In Case 0, the WDPs power consumption follows the water demand

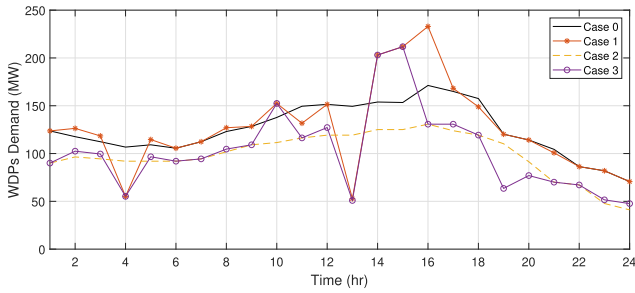


Fig. 6. Total power consumption by WDPs during a sample day.

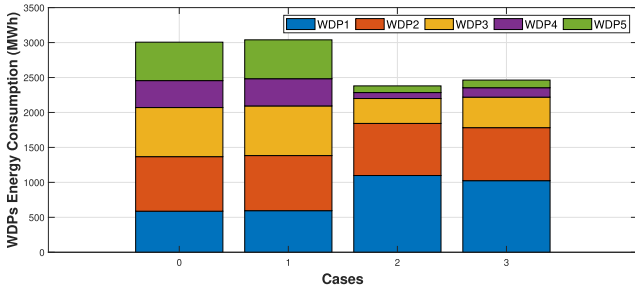


Fig. 7. Energy consumption by WDPs during a sample day.

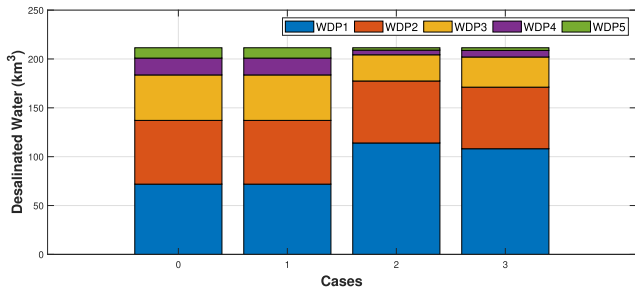


Fig. 8. Desalinated Water by WDPs during a sample day.

irrespective of the power system conditions. However, in Case 1, the WDPs power consumption flexibly changes in response to the power system's net load. For example, the net load is the lowest at hours 14:00-16:00. Therefore, the WDPs electric load during these hours in Case 1 is significantly higher than that of Case 0 schedule. This load is shifted from higher net load hours, e.g. 04:00 and 13:00. Because the WDP energy efficiency is a function of its operating point and water temperature, the change in WDP schedule has resulted in a change in WDP energy consumption (as seen in Fig. 7), even though the overall water production of each WDP throughout the day (as seen in Fig. 8) is fixed here. It is interesting to note that the total energy consumption of WDPs has slightly increased in Case 1 compared to Case 0 (see Fig. 7) although the overall operating cost has dropped (see Table II).

In Case 2, the WDPs power consumption also follows the water demand irrespective of the power system conditions (similar to Case 0). However, because in Case 2 the WDPs participation factors are not pre-determined, the WDPs scheduling favors the more efficient units. Hence, the WDPs power consumption in

TABLE II
NEXUS OPERATION RESULTS WITH THE NETWORK CONSTRAINTS

Cases	0	1	2	3
Cost (\$)	66190	63260.8	62173.3	59265.1
Cost of load reduction (\$)	0	1554.8	0	831.7
WDPs Consumption (MWh)	3005.6	3038.9	2380.2	2463.0
RES Curtail. (MWh)	1044.4	1036.0	1042.7	1036.0
Total net load peak (MW)	687.1	687.1	668.0	667.9
Desalinated water (m ³ /day)	211.5k	211.5k	211.5k	211.5k
% Cost reduction (Case 0)	0	4.43	-	-
% Cost reduction (Case 2)	-	-	0	4.68

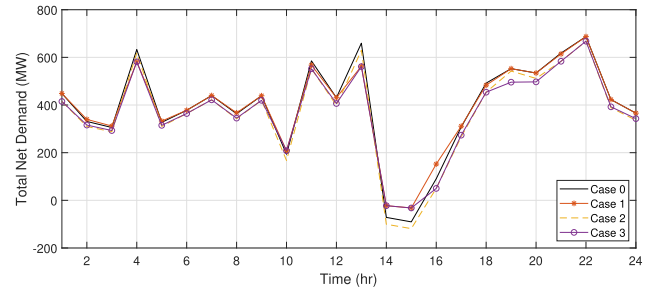


Fig. 9. The coupled net load profile of the nexus.

Case 2 is lower than that of Case 0. In Case 3, the WDPs power consumption fluctuates in response to the power system's net load. Since the net load is low at hours 14:00 and 15:00, the WDPs electric load increases during these hours above Case 2's schedule. This load is shifted from hours 13:00 and 19:00. It should be noted that the added load consumption (in MWh) in hours 14:00 and 15:00 is not necessarily equal to the reduced MWh consumption in hours 13:00 and 19:00. This is because the energy needed per unit water volume is not constant, as explained in Section II-B. Note that the water demand in both Cases 2 and 3 are the same (see new Fig. 8). Now, however, the total water production of each WDP is allowed to change from Case 2 to Case 3, given that the WDP participation factors are decision variables. Again, the savings in Case 3 are obtained by partially shifting the schedules of some WDPs from high power net-load periods to low power net-load periods, which has helped flatten the power net-load profile. Again, because the WDP energy efficiency is a function of its operating point and water temperature, the change in WDP schedule has resulted in a change in WDP energy consumption. It is also noted that the total energy consumption of WDPs has actually slightly increased in Case 3 compared to Case 2 (as seen in new Fig. 7), although the overall operating cost has dropped.

The coupled net load profile for the power-water system is shown in Fig. 9. This figure demonstrates that the peak net load has dropped by about 2.8% (comparing Case 3 to Case 0 net loads at hour 22:00). Note that the coupled net load was negative in hours 14:00 and 15:00 in all the cases. Hence, RES have been curtailed during these hours (as shown in Table II). It is worth noting that RES curtailment has been reduced in Cases 1 and 3 due to co-optimization. These results demonstrate the benefits of power-water coordination for a renewable-rich power system.

A breakdown of the scheduling results for the individual WDPs in Case 2 and Case 3 are presented in Figs. 10–13. These

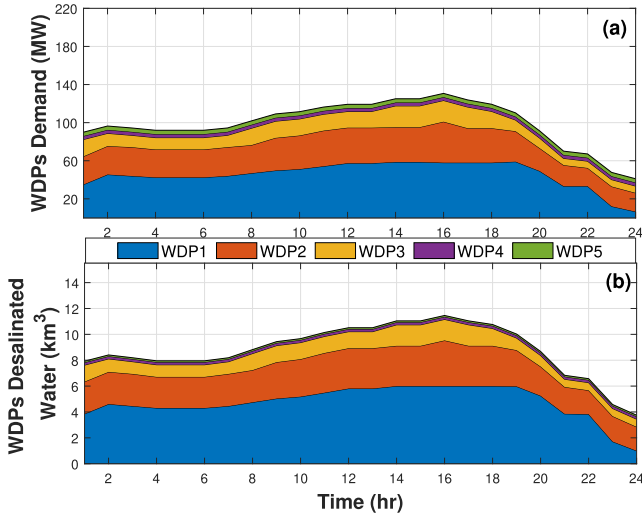


Fig. 10. (a) WDPs electric load, (b) permeate water volume – Case 2.

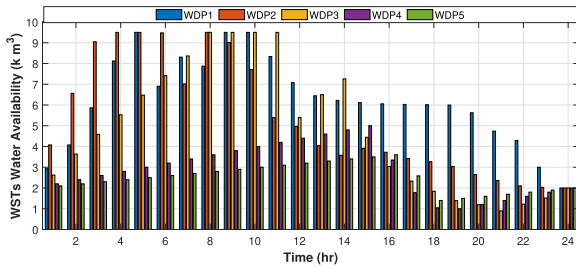


Fig. 11. Water volume available in the WSTs – Case 2.

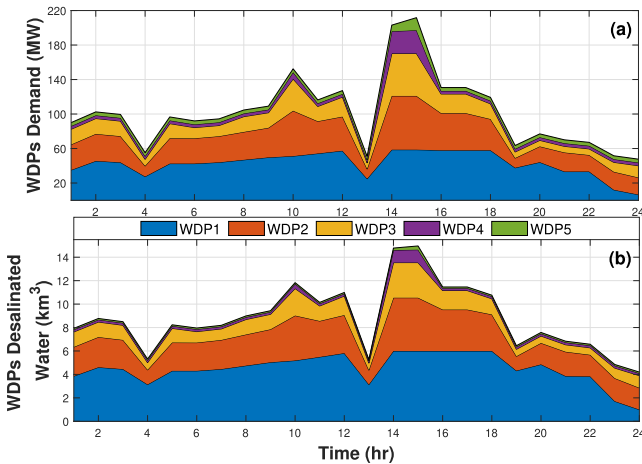


Fig. 12. (a) WDPs electric load, (b) permeate water volume – Case 3.

results show that WDP1 is mostly a base-load unit, WDP2 and WDP3 follow the net load profile, while WDP4 and WDP5 are operating at their minimum water desalination level most of the time. Note that during exceptionally light net loading conditions, these units were motivated to ramp up production, as seen in Fig. 10 during hours 14:00 and 15:00. The units are avoiding shutting down during low-activity periods due to their relatively high start-up and shut-down costs, which is typical

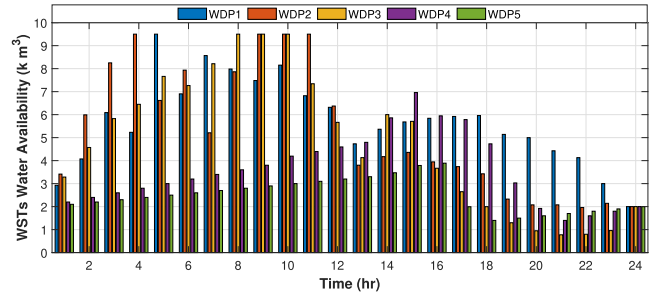


Fig. 13. Water volume available in the WSTs – Case 3.

TABLE III
NEXUS OPERATION RESULTS WITHOUT THE NETWORK CONSTRAINTS

Cases	2 (Without Network Constraints)	3 (Without Network Constraints)
Cost (\$)	57608.5	55662.5
Cost of load reduction (\$)	0	1072.6
WDPs Consumption (MWh)	2380.2	2673.1
RES Curtail. (MWh)	239.2	59.0
Total net load peak (MW)	668.0	659.1
Desalinated water (m ³ /day)	211.5k	211.5k
% Cost reduction (Case 2)	0	3.38

for RO desalination plants. The flexibility in WDP scheduling is possible due to the existence of storage tanks that store the desalinated water during water off-peak periods to serve the peak water demand. As shown in Fig. 13 for Case 3, the water levels available in the WSTs of the more efficient units (e.g., WDP1 and WDP2) reach the maximum WSTs capacity more often than that of the less efficient units. This demonstrates the tendency of the model to utilize the more efficient units to the highest possible extent.

2) *Operation Costs*: The total operation cost of the coupled power-water system for the four cases is shown in Table II. It is shown that the total operation cost of Case 3 is less than that of Case 1 by 6.3%. This demonstrates the benefits of considering the participation factors as decision variables. Comparing the operating cost of Case 3 to that of Case 2 demonstrates the benefit of simultaneous optimization in reducing the cost by 4.7%. Note that this cost reduction occurs despite the slight increase in the total WDP energy consumption, as shown in Fig. 7. The combined benefits of simultaneous optimization and having the WDP participation factors as decision variables are evident by comparing Case 3's cost to Case 0's, where the cost reduction is more than 10% and the energy consumption is significantly reduced. Note that the daily desalinated water volume is the same in all cases, as shown in Fig. 8.

To further study the benefits of the joint water-power operation, Case 2 and Case 3 are run ignoring the power network constraints, and the results are summarized in Table III. In Table III, renewable curtailment is slashed by 75% and net load peak is lowered by 1% due to the joint operation. These results indicate that joint operation can have different benefits at different power system's operating conditions, and the benefits are higher in systems without transmission congestion.

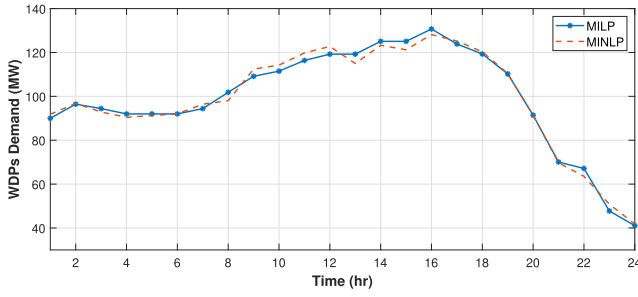


Fig. 14. Total power consumption by WDPs during a sample day.

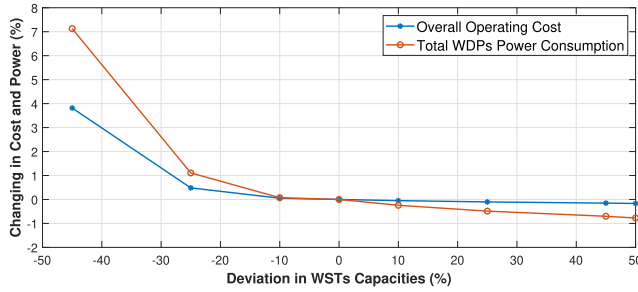


Fig. 15. Sensitivity analysis for different WST capacities – Case 3.

3) Comparisons of Linear and Non-Linear RO Plant Models:

In this exercise, it is aimed to compare the proposed linearized operation model of WDP's performance that is presented in Section III with the nonlinear one in Section II-A. The co-optimized water-power operational optimization is formulated as MILP and MINLP which are solved using CPLEX and BARON solvers, respectively, in a GAMS environment. The scheduling of WDP's energy consumption in Case 2 for both models is shown in Fig. 14. This figure shows that the schedules marginally differ. The MINLP model achieves slightly lower operating costs but takes considerably more computation time. The MINLP operating cost is only 0.03% less than that obtained by the MILP, but the MINLP computation time is more than 16 times higher. These results demonstrate that the proposed linearization gives comparatively accurate results at significantly lower computation time.

4) *Sensitivity Analysis:* A sensitivity analysis is carried out here to study the impacts of different WST capacities on the results of Case 3. More specifically, simulations are carried out after changing the WST capacities of all WDPs by plus and minus 10, 25, and 45%. In each case, the changes in system's total operating cost and WDPs power consumption are observed. The results of this sensitivity analysis are shown in Fig. 15. Evidently, increasing the WST capacities had little impact on the results, indicating that the original WST capacity constraints were not binding. On the other hand, reducing the WST capacities by 45% has resulted in a rise in the system's overall operating cost by almost 4%, indicating that the WST capacity constraints turned binding in many more hours than the original capacity constraints. In this case, the total WDP's power consumption has also increased by more than 7%. To better understand the reason behind this hike in power consumption, we show the

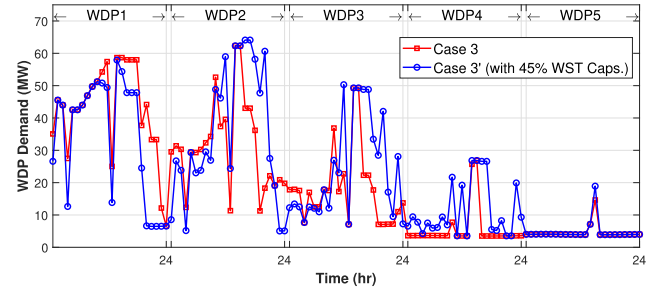


Fig. 16. WDP power consumption for different WST capacities – Case 3.

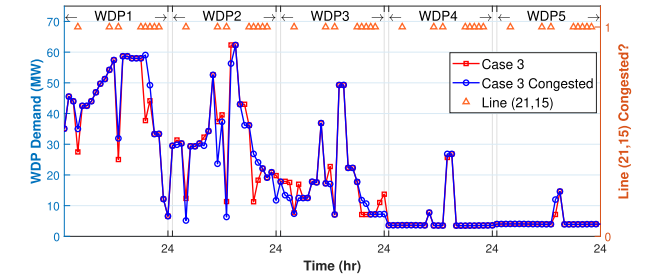


Fig. 17. WDP power consumption with line congestion – Case 3.

WDP power demand profiles for the original Case 3 and the constrained Case 3 in Fig. 16. These profiles suggest that the more constrained storage capacity case caused the less efficient units (WDP3, 4, and 5) to contribute more because the most efficient unit (WDP1) is bound by its own WST capacity.

Furthermore, to have a deeper look at how the joint operation of water and power systems helps mitigate congestion, an additional simulation is carried out at which the power flow limit of line 21–15 is reduced from 400 to 250 MW. A comparison between the less congested case and the more congested case is shown in Fig. 17. Note that WDP2 is located at bus 15, which is the receiving end of this line. During hours of congestion (e.g. hours 04:00, 11:00, and 13:00), WDP2 tends to reduce its power consumption to help relieve the congested line 21–15. This water demand is moved to other WDPs (e.g. WDP1 in hours 04:00 and 11:00). These responses from WDPs are also combined with increases in the output power of thermal generator unit 6, which is also located at bus 15, during hours of congestion (the output of generator 6 is not shown in Fig. 17). During uncoordinated operation, congestion in line 21–15 relies primarily on more significant increases in the output power of generator 6. This demonstrates the benefits of joint operation to mitigate line congestion.

B. Study 2: Stochastic Operation Model

This study presents the numerical results of Case 4 with risk-neutral and risk-aversion optimization strategies. In the risk-neutral strategy, the linear SP model for Case 4 is solved with $\varphi = 0$. For comparison, the average realized costs (ARC) are obtained for Cases 3 and 4. To obtain the ARC for each case, the schedules obtained from the corresponding optimization are used. Then, the realized cost of operation considering each of

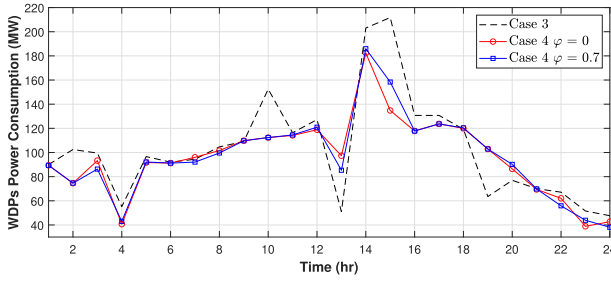


Fig. 18. Total power consumption by WDPs during a sample day.

TABLE IV
COMPARISON OF DETERMINISTIC AND STOCHASTIC MODELS

Cases	3	4 ($\varphi = 0$)	4 ($\varphi = 0.7$)
Expected Cost (\$)	95017.3	64053.2	64076.2
CVaR (\$)	-	82354.2	65033.3
Cost of load reduction (\$)	1153.2	729.3	817.3
WDPs Consumption (MWh)	2479.13	2311.95	2317.9
Avg. RES Curtail. (MWh)	2059.4	1771.5	1756.6
Avg. net load peak (MW)	685.7	662.9	656.7
Desalinated water (m^3/day)	211.5k	211.5k	211.5k

the 27 scenarios mentioned in subsection V-B is obtained. Then, ARC is obtained by averaging all the per-scenario costs. The ARC obtained using the deterministic schedules is \$95 k while that obtained using the risk-neutral strategy is \$64.1 k. That is, the latter results in about 32.5% savings. In addition, it is found that the WDPs energy consumption fluctuates less in Case 4 compared to that of Case 3, as observed in Fig. 18. This is because the WDPs energy consumption in Case 3 is based on a single scenario that is assumed to be certain. However, WDPs consumption of Case 4 is based on multiple scenarios which are uncertain. The results summarized in Table IV highlight more benefits for the stochastic model over the deterministic model. The stochastic model results in lower ARC, lower expected cost for load reduction, lower expected WDPs consumption, lower expected RES curtailment, and lower expected net load peak.

In the risk-aversion strategy, the linear SP model is solved for Case 4 with $\varphi = 0.7$. The results show that the corresponding CVaR is equal to \$65.0 k, which represents 21% reduction from the CVaR obtained using the risk-neutral strategy. This is achieved at the expense of a slight increase in the expected operation cost (expected operation cost with $\varphi = 0.7$ is \$64.08 k while that with $\varphi = 0$ is \$64.05 k). Moreover, the WDPs schedules when $\varphi = 0.7$ tend to be more responsive to the extreme conditions at hours 13:00 and 15:00 in order to decrease risks, as seen in Fig. 18. Detailed numerical results are shown in Table IV as comparisons between the deterministic Case (Case 3) and the stochastic Cases (Case 4 ($\varphi = 0$) and Case 4 ($\varphi = 0.7$)).

VII. CONCLUSION

This paper develops a novel unit commitment model to integrate the flexibility of RO water desalination plants to the DA power system operation. It considers a centralized entity that is responsible for simultaneously optimizing the scheduling

of the power system generation resources and RO plants. The power system operator receives data from both the power plants and the water desalination plants. The power system operator shares the characteristics of the thermal generation units and the forecasted electric load and RES generation. The water desalination operator shares the RO plants characteristics, the forecasted water demands, and the seawater temperatures. The power system operator co-optimizes the operating schedules of both systems aiming to minimize the coupled operating costs of the power-water nexus. The simulation results show that the daily operating cost of the coupled system is decreased, the energy consumed by the RO plants are reduced, the peak electricity demand is decreased, and the utilization of RES surplus generation is increased.

REFERENCES

- [1] K. Oikonomou, M. Parvania, and S. Burian, "Integrating water distribution energy flexibility in power systems operation," in *Proc. IEEE Power Energy Soc. Gen. Meeting*, 2017, pp. 1–5.
- [2] S. Shu, D. Zhang, S. Liu, M. Zhao, Y. Yuan, and H. Zhao, "Power saving in water supply system with pump operation optimization," in *Proc. Asia-Pacific Power Energy Eng. Conf.*, 2010, pp. 1–4.
- [3] A. Siddiqi and L. D. Anadon, "The water-energy nexus in Middle East and North Africa," *Energy Policy*, vol. 39, no. 8, pp. 4529–4540, Aug. 2011.
- [4] I. Karakitsios, A. Dimeas, and N. Hatziaargyriou, "Optimal management of the desalination system demand in non-interconnected islands," *Energies*, vol. 13, no. 15, Jan. 2020, Art. no. 4021.
- [5] I. M. Almanjahie, Z. Chikr-Elmezzouar, and A. Bachir, "Modeling and forecasting the household water consumption in Saudi Arabia," *Appl. Ecol. Environ. Res.*, vol. 17, no. 1, pp. 1299–1309, Jan. 2019.
- [6] Electricity and Water Authority of Bahrain, "Yearly total water production / average storage report - 2018," 2018. Accessed: Feb. 16, 2022. [Online]. Available: <https://www.ewa.bh/en/Network/Water/Statistics/2018/Pages/WaterREYearly-Production---Storage-Report---2018.aspx>
- [7] National Centre for Statistics and information of Oman, "Monthly desalinated water production in cubic meters - 2020," 2020. Accessed: Feb. 16, 2022. [Online]. Available: <https://data.gov.om/OMWTR2016/water>
- [8] Global Water Intelligence, "The IDA water security handbook," 2021. [Online]. Available: <https://www.globalwaterintel.com/products-and-services/market-research-reports/ida-water-security-handbook>
- [9] H. T. El-Dessouky and H. M. Ettouney, *Fundamentals of Salt Water Desalination*. Amsterdam, The Netherlands: Elsevier, 2002.
- [10] Invest Saudi, "Water - Desalination detailed value proposition," 2018. Accessed: Nov. 10, 2020. [Online]. Available: https://investsaudi.sa/media/1334/water-brochure_updated.pdf
- [11] A. Santhosh, A. M. Farid, and K. Youcef-Toumi, "The impact of storage facility capacity and ramping capabilities on the supply side economic dispatch of the energy-water nexus," *Energy*, vol. 66, pp. 363–377, Mar. 2014.
- [12] A. Santhosh, A. Farid, and K. Youcef-Toumi, "The impact of storage facilities on the simultaneous economic dispatch of power and water networks limited by ramping constraints," in *Proc. IEEE Int. Conf. Ind. Technol.*, 2013, pp. 922–927.
- [13] A. Santhosh, A. M. Farid, and K. Youcef-Toumi, "Optimal network flow for the supply side of the energy-water nexus," in *Proc. IEEE Int. Workshop Intelligent Energy Syst.*, 2013, pp. 155–160.
- [14] D. Kong and K. E. Holbert, "Solar thermal electricity generation and desalination in the Southwestern United States," in *Proc. North Amer. Power Symp.*, 2010, pp. 1–8.
- [15] T. Novosel *et al.*, "Integration of renewables and reverse osmosis desalination-case study for the Jordanian energy system with a high share of wind and photovoltaics," *Energy*, vol. 92, pp. 270–278, Dec. 2015.
- [16] L. House, "Water supply related electricity demand in California," Demand Response Research Center, U.S. Department of Energy, Contract, no. 500-03-026, and DE-AC02-05CH11231, pp. 1–77, 2006. Accessed: Nov. 01, 2021. [Online]. Available: <https://waterandenergyconsulting.com/WEDemandReport.pdf>
- [17] R. Segurado, J. Madeira, M. Costa, N. Duić, and M. G. Carvalho, "Optimization of a wind powered desalination and pumped hydro storage system," *Appl. Energy*, vol. 177, pp. 487–499, Sep. 2016.

- [18] K. Oikonomou and M. Parvania, "Optimal coordination of water distribution energy flexibility with power systems operation," *IEEE Trans. Smart Grid*, vol. 10, no. 1, pp. 1101–1110, Jan. 2019.
- [19] K. Oikonomou and M. Parvania, "Optimal coordinated operation of interdependent power and water distribution systems," *IEEE Trans. Smart Grid*, vol. 11, no. 6, pp. 4784–4794, Nov. 2020.
- [20] C. Li, Y. Yao, K. Xie, B. Hu, and T. Niu, "Integrated electrical, heating, and water distribution system to accommodate wind power," *IEEE Trans. Sustain. Energy*, vol. 12, no. 2, pp. 1100–1114, Apr. 2021.
- [21] L. Thompson, K. Song, A. Lekov, and A. McKane, "Automated demand response opportunities in wastewater treatment facilities," Lawrence Berkeley Nat. Lab. (LBNL), Berkeley, CA, USA, DOE Contract Number: DE-AC02-05CH11231, Tech. Rep. LBNL-1244E, Nov. 2008.
- [22] M. Elsir and A. T. Al-Awami, "Coordinating the operation of water desalination and a renewables-rich electric power grid," in *Proc. IEEE Power Energy Soc. Gen. Meeting*, 2020, pp. 1–5.
- [23] F. Mohammadi, M. Sahraei-Ardakani, Y. M. Al-Abdullah, and G. T. Heydt, "Coordinated scheduling of power generation and water desalination units," *IEEE Trans. Power Syst.*, vol. 34, no. 5, pp. 3657–3666, Sep. 2019.
- [24] A. A. Atia and V. Fthenakis, "Active-salinity-control reverse osmosis desalination as a flexible load resource," *Desalination*, vol. 468, Oct. 2019, Art. no. 114062.
- [25] K. Oikonomou, M. Parvania, and R. Khatami, "Optimal demand response scheduling for water distribution systems," *IEEE Trans. Ind. Inform.*, vol. 14, no. 11, pp. 5112–5122, Nov. 2018.
- [26] K. Oikonomou and M. Parvania, "Optimal participation of water desalination plants in electricity demand response and regulation markets," *IEEE Syst. J.*, vol. 14, no. 3, pp. 3729–3739, Sep. 2020.
- [27] K. Oikonomou and M. Parvania, "Deploying water treatment energy flexibility in power distribution systems operation," in *Proc. IEEE Power Energy Soc. Innov. Smart Grid Technol. Conf.*, 2020, pp. 1–5.
- [28] Solutions, DOW Water, "FilmTec reverse osmosis membranes," vol. 399, no. 609-00071, pp. 1–181, 2010. [Online]. Available: <https://anq.org.mx/pqta/pdf/Respaldo/MANUAL%20OSMOSIS%20INVERSA.pdf>
- [29] I. S. Al-Mutaz and M. A. Al-Ghunaimi, "Performance of reverse osmosis units at high temperatures," in *Proc. IDA World Congr. Desalination Water Reuse*, 2001, pp. 26–31.
- [30] P. Kall, S. W. Wallace, and P. Kall, *Stochastic Programming*. Berlin, Germany: Springer, 1994.
- [31] A. T. Al-Awami and M. A. El-Sharkawi, "Coordinated trading of wind and thermal energy," *IEEE Trans. Sustain. Energy*, vol. 2, no. 3, pp. 277–287, Jul. 2011.
- [32] A. T. Al-Awami and E. Sortomme, "Coordinating vehicle-to-grid services with energy trading," *IEEE Trans. Smart Grid*, vol. 3, no. 1, pp. 453–462, Mar. 2012.
- [33] F. Angizeh and M. Parvania, "Stochastic risk-based flexibility scheduling for large customers with onsite solar generation," *IET Renewable Power Gener.*, vol. 13, no. 14, pp. 2705–2714, 2019.
- [34] S. Bahramirad, W. Reder, and A. Khodaei, "Reliability-constrained optimal sizing of energy storage system in a microgrid," *IEEE Trans. Smart Grid*, vol. 3, no. 4, pp. 2056–2062, Dec. 2012.
- [35] M. Parvania, M. Fotuhi-Firuzabad, and M. Shahidehpour, "ISO's optimal strategies for scheduling the hourly demand response in day-ahead markets," *IEEE Trans. Power Syst.*, vol. 29, no. 6, pp. 2636–2645, Nov. 2014.
- [36] C. Ordoudis, P. Pinson, J. M. Morales, and M. Zugno, "An updated version of the IEEE RTS 24-bus system for electricity market and power system operation studies," 2016. [Online]. Available: https://backend.orbit.dtu.dk/ws/portalfiles/portal/120568114/An_Updated_Version_of_the_IEEE_RTS_24Bus_System_for_Electricity_Market_an....pdf
- [37] M. Elsir, "Online materials," 2021. Accessed: Nov. 18, 2021. [Online]. Available: https://pdfhost.io/v/NBVDUtlk_Risk_Based_Operation_Coordination_of_Water_Desalination_and_Renewable_Rich_Power_Systems_DATA

Mohamed Elsir received the B.Sc. degree in nuclear engineering from the Sudan University of Science and Technology, Khartoum, Sudan, in 2015, and the M.Sc. degree in electrical engineering from the King Fahd University of Petroleum and Minerals, Dhahran, Saudi Arabia, in 2020. He is currently working toward the Ph.D. with the Department of Electrical Engineering and Computer Science, Khalifa University of Science and Technology, Abu Dhabi, UAE. His research interests include integrating energy-water nexus, demand response applications, energy storage systems, and renewable generation resources into the smart grid.

Ali T. Al-Awami (Senior Member, IEEE) received the B.Sc. and M.Sc. degrees in electrical engineering from the King Fahd University of Petroleum and Minerals (KFUPM), Dhahran, Saudi Arabia, in 2000 and 2005, respectively, and the Ph.D. degree in electrical engineering from the University of Washington, Seattle, WA, USA, in 2010. He is currently an Associate Professor with the Department of Electrical Engineering, KFUPM. His research interests include power system operation and optimization, and the integration of hybrid energy systems. His industrial experiences include working for the Saudi Electricity Company, Saudi Arabia and Bonneville Power Administration, USA. He was the recipient of several best conference article and best poster awards from a number of conferences, including the IEEE Power and Energy Society General Meeting in 2009, IEEE Clemson University Power Systems Conference in 2016, and Saudi Smart Grid Conferences (SASG) in 2013 and 2018.

Mohamed A. Antar received the Ph.D. degree from the King Fahd University of Petroleum and Minerals (KFUPM), Dhahran, Saudi Arabia, in 1996. He is currently a Professor with the Department of Mechanical Engineering, KFUPM. His research interests include thermofluid systems, renewable energy, and water desalination by thermal and membrane systems. He has established Water Desalination Lab, KFUPM. He was the recipient of teaching, advising instruction technology, and research awards at KFUPM. He has along list of papers and patents in these areas.

Konstantinos Oikonomou (Member, IEEE) received the B.Sc. degree in physics from the University of Patras, Patras, Greece, in 2009, the first M.Sc. degree in energy engineering from the University of Durham, Durham, U.K., in 2011, the second M.Sc. degree in electrical engineering from the George Washington University, Washington, DC, USA, in 2015, and the Ph.D. degree in electrical engineering from The University of Utah, Salt Lake City, UT, USA, in 2019. He is currently an Electrical Engineer in the energy and environment directorate with Pacific Northwest National Laboratory, where he studies the operation, optimization, economics, and resilience of interdependent critical infrastructures.

His research interests include designing electricity market participation frameworks for active power and water distribution networks and developing physics-based models to capture dynamic multiscale interactions, feedbacks, and flexibility across power, water, gas, and transportation systems.

Masood Parvania (Senior Member, IEEE) is currently an Associate Professor and the Associate Chair for research and advancement with the Department of Electrical and Computer Engineering, University of Utah, Salt Lake City, UT, USA. His research interests include the operation, economics, and resilience of power and energy systems, and also modeling and operation of interdependent critical infrastructures. Dr. Parvania is an Associate Editor for the IEEE TRANSACTIONS ON POWER SYSTEMS, IEEE TRANSACTIONS ON SMART GRID, and IEEE TRANSACTIONS ON SUSTAINABLE ENERGY. He is currently the Chair of the IEEE Power and Energy Society (PES) Utah Chapter, Chair of the IEEE PES Bulk Power System Operation Subcommittee, and Chair of the IEEE PES Reliability, Risk and Probability Application (RRPA) Subcommittee.